Customer Shopping Behavior Analysis

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1. Project Overview

This project explores customer shopping behavior using transaction data collected from approximately 3,900 purchases across multiple product categories. The purpose of this analysis is to identify trends in customer spending, purchasing habits, product preferences, and subscription usage. The insights gained from this analysis can help businesses make informed decisions related to marketing, customer retention, and sales strategies.

Technical Skills Used:

- **SQL (MySQL):** Complex queries using JOINs, GROUP BY, subqueries, CASE statements, and customer segmentation logic
- **Python (Pandas):** Data cleaning, missing value handling, feature engineering, and exploratory data analysis
- **Data Cleaning:** Handling missing values, standardizing columns, and removing unneeded data
- **Power BI:** Interactive dashboard creation, KPIs, filters, and data visualization
- **Database Management:** Loading cleaned datasets into MySQL for analysis
- **Business Analysis:** Translating data insights into marketing and retention recommendations

Key Takeaways:

This project uncovered which customer groups contribute the most revenue, highlighted products that rely heavily on discounts, and showed that repeat customers are more likely to become subscribers. These findings give businesses clearer direction on where to focus marketing efforts, how to adjust discount strategies, and how to strengthen long term customer retention.

2. Dataset Summary

- **Total Rows:** 3,900
- **Total Columns:** 18

Key Variables Included:

- Customer demographics such as age, gender, location, and subscription status
- Purchase information including item purchased, product category, purchase amount, season, size, and color
- Shopping behavior indicators such as discount usage, promo code usage, number of previous purchases, purchase frequency, shipping type, and customer review ratings

Data Quality Notes:

- A total of 37 missing values were identified in the Review Rating column, which required cleaning during data preparation.

3. Exploratory Data Analysis Using Python

Data preparation and exploration were performed using Python and the pandas library.

- **Data Import:**
  - The dataset was loaded into a pandas DataFrame for analysis.
- **Initial Exploration:**
  - `df.info()` was used to review data types and structure.
  - `df.describe()` was used to examine summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

- **Missing Data Handling:**
  - Missing values in the Review Rating column were filled using the median rating for each product category to maintain realistic customer feedback patterns.
- **Column Standardization:**
  - Column names were converted to snake_case to improve readability and consistency.

- **Feature Engineering:**
    - An age_group column was created by grouping customers into age ranges.
    - A purchase frequency feature was derived to better represent buying behavior.
- **Data Consistency Check:**
    - The discount_applied and promo_code_used columns were found to be redundant.
    - The promo_code_used column was removed to simplify the dataset.
- **Database Integration:**
    - The cleaned dataset was loaded into a MySQL database to support SQL-based analysis.

4. Data Analysis Using SQL (Business Transactions)

SQL queries were executed in MySQL to answer key business questions:

1. **Revenue by Gender**
    - Compared total revenue generated by male and female customers.

gender	revenue
Male	157890
Female	75191

2. **High-Spending Discount Users**
    - Identified customers who used discounts but still spent more than the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67
35	91
37	69
40	60
41	76
43	100
44	69
55	94
57	73
58	64
60	79
62	68

3. **Top Five Products by Review Rating**

    ○ Determined products with the highest average customer ratings.

item_purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

4. **Shipping Type Comparison**
    - Compared average purchase amounts between Standard and Express shipping options.

shipping_type	ROUND(AVG(purchase_amount),2)
Express	60.48
Standard	58.46

5. **Subscribers vs. Non-Subscribers**
    - Analyzed differences in average spending and total revenue based on subscription status.

subscription_status	total_customers	avg_spend 2	total_revenue 1
No	2847	59.87	170436
Yes	1053	59.49	62645

6. **Discount-Dependent Products**
    - Identified the five products with the highest percentage of discounted purchases.

item_purchased	discount_rate 1
Hat	50.00
Coat	49.00
Sneakers	49.00
Sweater	48.00
Pants	47.00

7. **Customer Segmentation**
    - Classified customers as New, Returning, or Loyal based on purchase history.

customer_segment	Number of Customer
Loyal	3116
Returning	701
New	83

8. **Top Three Products per Category**

- Identified the most frequently purchased products within each category.

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

9. **Repeat Buyers and Subscription Behavior**
    - Evaluated whether customers with more than five purchases were more likely to subscribe.

subscription_status	repeat_buyers
Yes	958
No	2518

10. **Revenue by Age Group**
    - Calculated total revenue contributions across different age groups.

age_group	total_revenue 1
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

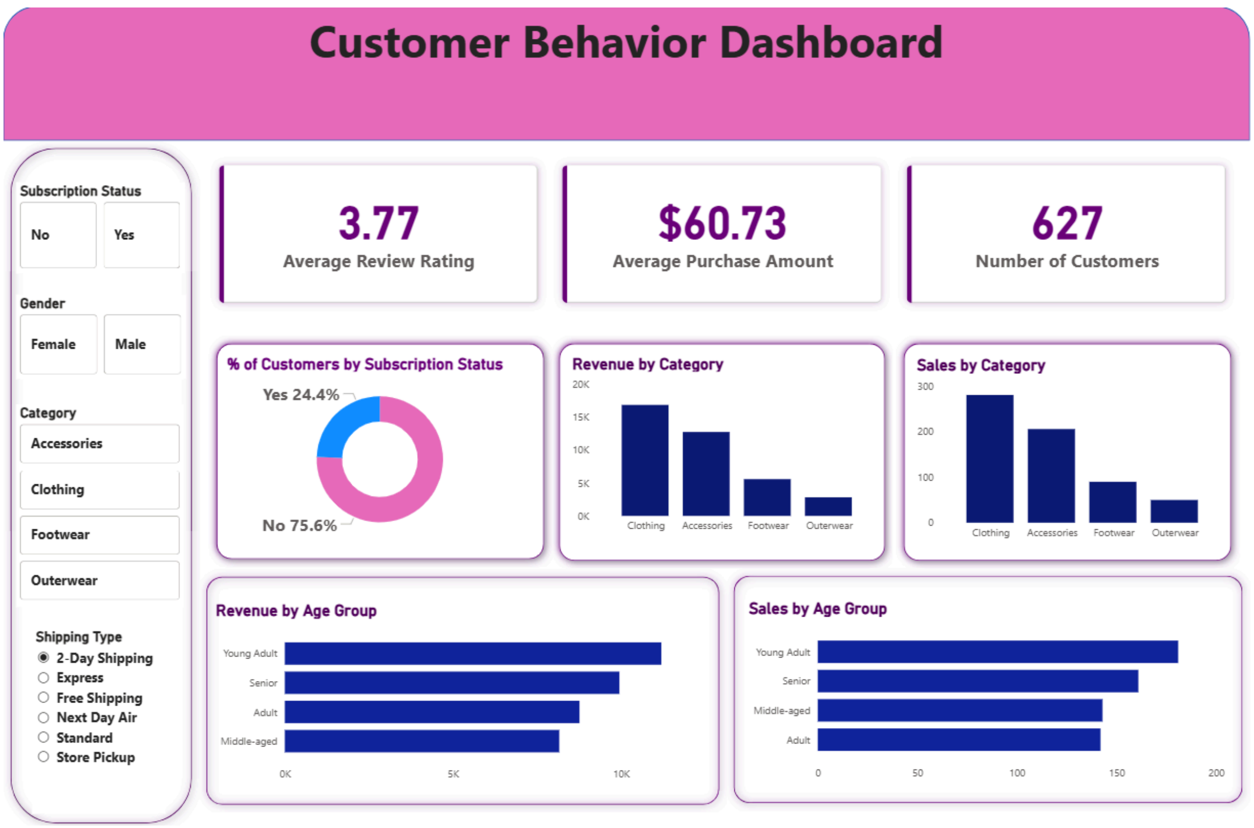
5. Dashboard in Power BI

An interactive dashboard was developed in Power BI to visually present the analysis results.

- Key metrics displayed include:

    ◦ Average review rating
    ◦ Average purchase amount
    ◦ Total number of customers
- Additional visuals highlight:
    ◦ Customer segmentation
    ◦ Revenue distribution
    ◦ Purchasing trends across demographics and behaviors

The dashboard allows users to interact with the data and explore insights dynamically.



6. Business Recommendations

Based on the analysis, the following recommendations are suggested:

- **Increase Subscription Enrollment:**
    - Offer exclusive benefits or incentives to encourage customers to subscribe.
- **Strengthen Customer Loyalty Programs:**
    - Reward repeat customers to move them into the Loyal customer segment.
- **Review Discount Strategies:**
    - Ensure discounts increase sales while maintaining healthy profit margins.
- **Improve Product Promotion:**
    - Highlight top rated and best selling products in marketing campaigns.
- **Target High-Value Segments:**
    - Focus marketing efforts on high-revenue age groups and customers who prefer express shipping.